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Journal of Mathematical Biology

ISSN 0303-6812

J. Math. Biol.

DOI 10.1007/s00285-017-1204-8



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Multi-scale modeling of APC and β -catenin regulation in the human colonic crypt

Brooks Emerick¹ · Gilberto Schleiniger² · Bruce M. Boman^{3,4}

Received: 20 January 2017 / Revised: 22 December 2017
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Abstract Stem cell renewal and differentiation in the human colonic crypt are linked to the Wnt/ β -catenin pathway. The spatial balance of Wnt factors in proliferative cells within the crypt maintain an appropriate level of cellular reproduction needed for normal crypt homeostasis. Mutational events at the gene level are responsible for deregulating the balance of Wnt factors along the crypt, causing an overpopulation of proliferative cells, a loss of structure of the crypt domain, and the initiation of colorectal carcinomas. We formulate a PDE model describing cell movement and reproduction in a static crypt domain. We consider a single cell population whose proliferative capabilities are determined by stemness, a quantity defined by intracellular levels of adenomatous polyposis coli (APC) scaffold protein and β -catenin. We fit APC regulation parameters to biological data that describe normal protein gradients in the crypt. We also fit cell movement and protein flux parameters to normal crypt

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s00285-017-1204-8>) contains supplementary material, which is available to authorized users.

✉ Brooks Emerick
bemerick@kutztown.edu

Gilberto Schleiniger
schleini@udel.edu

Bruce M. Boman
BrBoman@ChristianaCare.org

- ¹ Department of Mathematics, Kutztown University, Kutztown, PA 19530, USA
- ² Department of Mathematical Sciences, University of Delaware, Newark, DE 19711, USA
- ³ Department of Biological Sciences, University of Delaware, Newark, DE 19711, USA
- ⁴ Center for Translational Cancer Research, Helen F. Graham Cancer Center and Research Institute, Newark, DE 19713, USA

characteristics such as renewal time, total cell count, and proportion of proliferating cells. The model is used to investigate abnormal crypt dynamics when subjected to a diminished APC gradient, a scenario synonymous to mutations in the *APC* gene. We find that a 25% decrease in APC synthesis leads to a fraction of 0.88 proliferative, which is reflective of normal-appearing FAP crypts. A 50% drop in APC activity yields a fully proliferative crypt showing a doubling of the level of stemness, which characterizes the initial stages of colorectal cancer development. A sensitivity analysis of APC regulation parameters shows the perturbation of factors that is required to restore crypt dynamics to normal in the case of *APC* mutations.

Keywords Colonic crypt · APC regulation · Wnt pathway · Colorectal cancer · Multi-scale model

Mathematics Subject Classification 92C37 · 92C45 · 92C17 · 92C40 · 92D20

1 Introduction

A cell signaling pathway is an intracellular biochemical process that is often initiated by an extracellular stimulus. The signal transduction is characterized by a cascade of biochemical reactions that acts as a complex communication network and induces a physiological response important in embryonic development, cell function, and tissue maintenance. This basic, yet complex, process is key to normal cell function; however, abnormal regulation of the pathway components can lead to developmental defects as well as cancerous states (Weinberg 2007; Nusse 2005; Clevers and Nusse 2012). The Wnt signaling pathway controls biological phenomena throughout the developmental and adult stages of all animals. The hallmark of the Wnt pathway is the regulation of cytoplasmic β -catenin protein concentrations, whose activity modulates cellular proliferation, differentiation, and cell-to-cell adhesion (Clevers and Nusse 2012; Lloyd-Lewis et al. 2013; Kikuchi and Yamamoto 2007). The stabilization of this protein leads to normal cell and tissue function while perturbations can cause human degenerative diseases as well as cancer. Therefore, it is essential to understand the underlying mechanisms responsible for this dysfunctional pathway.

The canonical Wnt signaling pathway plays an important role in regulating the amount of β -catenin by acting as a switch. In the absence of Wnt, a destruction complex made up of adenomatous polyposis coli (APC) scaffold protein, a scaffolding protein called Axin, and the enzyme GSK-3 β phosphorylates β -catenin, which quickly leads to proteolysis (Weinberg 2007; Aberle et al. 1997; Latres et al. 1999; Liu et al. 1999; MacDonald et al. 2009). In the presence of Wnt, the destruction complex is deactivated by the intermediate protein Dishevelled (Dsh) and accumulation of β -catenin occurs within the cytoplasm. β -catenin binds with transcription factors (TCF/LEF) to drive the expression of a variety of genes, particularly those responsible for cell proliferation (Weinberg 2007; Vries et al. 2010). Essentially, a bound Wnt molecule systematically inhibits the destruction of β -catenin allowing it to accumulate and promote cellular proliferation.

Wnt induced β -catenin activity is directly regulated by the APC protein. APC interacts with Axin through three binding motifs that also bind with β -catenin (Clevers and Nusse 2012). Furthermore, APC plays a role in spindle orientation and microtubule function during mitosis (Boman and Fields 2013), which means it is an active participant in cell differentiation. It is also well known that the APC protein loses its function through a series of mutations (Lloyd-Lewis et al. 2013; Boman and Fields 2013; Cho et al. 2006; Wodarz and Komarova 2005). The APC protein is encoded by the *APC* tumor suppressor gene, which is responsible for suppressing the growth of damaged or mutated cells (Wodarz and Komarova 2005). Tumor suppressor genes are deactivated when both the maternal and paternal copies of the gene become mutated (Cho et al. 2006; Wodarz and Komarova 2005). The systematic loss of function of the APC protein leads to the expression of a truncated APC protein lacking the specific binding site to the destruction complex (Fearnhead et al. 2001; Gaspar and Fodde 2004; Albuquerque et al. 2002). This allows β -catenin to accumulate in the cytosol, which mimics the effect of Wnt signaling, and ultimately contributes to tumorigenesis (Cho et al. 2006; Wodarz and Komarova 2005). Studies have determined that this process characterizes the initial stages of colorectal cancer development, but many of the APC protein's specific molecular activities remain uncertain.

Understanding the regulation of β -catenin and APC via Wnt is especially important in the colon. The epithelium of the large intestine contains approximately 10^7 finger-like invaginations known as colonic crypts (see Fig. 1). Approximately 3000 cells make up the crypt, with about 20 stem cells residing at the base (Potten and Loeffler 1990; Nishimura et al. 2003; Huang et al. 2009; Boman and Huang 2008). Stem cells can divide asymmetrically into one stem and one non-stem cell or they can divide symmetrically into two stem cells or two non-stem cells (Ro and Rannala 2001; Molofsky et al. 2004). Stem cells divide, on average, asymmetrically, thereby generating a stem cell population and transient amplifying (TA) cell population (Boman et al. 2007). Stem cells replenish themselves and are extremely long-lived. TA cells migrate up the crypt while dividing and are eventually extruded into the intestinal lumen (Fig. 1). As they migrate, they also mature along specific cell lineages including endocrine, absorptive, and goblet cells (Cheng and Leblond 1974; Shamsuddin et al. 1982). In essence, the progeny of the stem cells make up the epithelial lining of the colon, which is renewed every 3–5 days.

In addition to the phenotypical layout of the crypt, specific biochemical profiles are observed. Wnt ligands are produced near the crypt base by stromal cells in the extracellular matrix (Weinberg 2007; Holcombe et al. 2002; Gregorieff et al. 2005). This contributes to a gradient of Wnt factors along the major axis of the crypt (see Fig. 1). There is evidence that continual stimulation of Wnt activity in the crypt bottom is key for stem cell maintenance, normal TA cell proliferation, and crypt homeostasis (Gregorieff et al. 2005; Fevr et al. 2007). Furthermore, inverse gradients of intracellular β -catenin and APC are observed (Boman and Fields 2013). In experimental results conducted by Boman et al. and Kosinski et al., the amount of intracellular APC increases nearly fivefold as measured from the bottom of the crypt to the top (Zhang et al. 2010; Kosinski et al. 2007). Although an increase in APC is observed in cells as they migrate up the crypt (Midgley et al. 1997; Miyashiro et al. 1995; Smith et al. 1993; Umar et al. 2005; Reinacher-Schick and Gumbiner 2001;

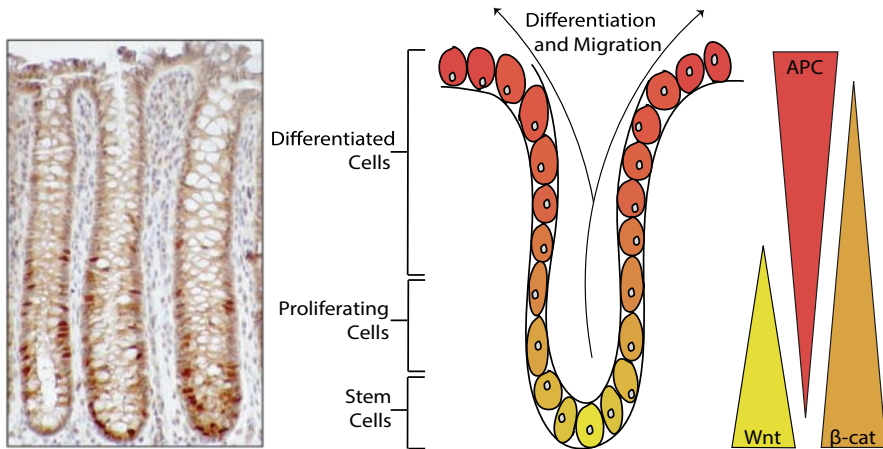


Fig. 1 (Left) A picture of a normal colonic crypt with immunohistochemical staining for Aurora B Kinase, which is an indicator of proliferative cells (Zhang et al. 2010). The darker color cells are more stem-like. (Right) A schematic illustration of the phenotypical layout and biochemical distribution of the crypt (color figure online)

Näthke et al. 1996; Senda et al. 1996; Maltzman et al. 1996; Wang et al. 2009), little is known about the molecular regulatory mechanism that generates this gradient; however, cellular proliferation appears to drive production of APC (Boman and Fields 2013; Fagman et al. 2003). Other studies suggest that promoter methylation and specific transcription factors such as p53, USF1, and USF2 may regulate the expression of APC (Jaiswal and Narayan 1998, 2001a, b; Jaiswal et al. 2006; Narayan and Jaiswal 1997; Virmani et al. 2001). Studies show that Wnt signaling components and APC are necessary for cell division (Boman and Fields 2013). It is thought that the inverse gradients create a zone of rapid proliferation approximately one-third of the way up the crypt that drives most of cell movement and crypt formation (Boman and Fields 2013). The inverse profiles of Wnt factors and intracellular APC proteins are responsible for position-dependent rates of cell proliferation, differentiation, and apoptosis. Therefore, understanding the pathway components is essential to understanding mechanisms that maintain healthy colonic crypt dynamics and organization as well as those that underlie colorectal cancer initiation.

About 80% of all sporadic colon carcinomas are initially characterized by an accumulation of β -catenin inside the cell (Weinberg 2007; Wood et al. 2007; Muzny et al. 2012). This imbalance is due to the two-hit mutation of the APC gene, resulting in a loss of function of the APC protein. Early findings suggest that proliferative properties of the crypt become dysregulated, which was shown using a cell labeling index (LI) along crypt height (Deschner et al. 1963; Bleiberg et al. 1972; Iwana et al. 1977; Potten et al. 1992) (see Fig. 2). LI measures the fraction of cells in the active cell proliferation phase, which normally results in a skewed bell shaped curve against crypt level attaining its maximum near the base of the crypt. The first hit mutation, i.e. germline mutant, is studied in most hereditary cases of familial adenomatous polyposis (FAP). FAP crypts are normal-appearing and not yet dysplastic but show a shift in the cell LI, maximizing at approximately cell level 20 where normal crypts maximize at level

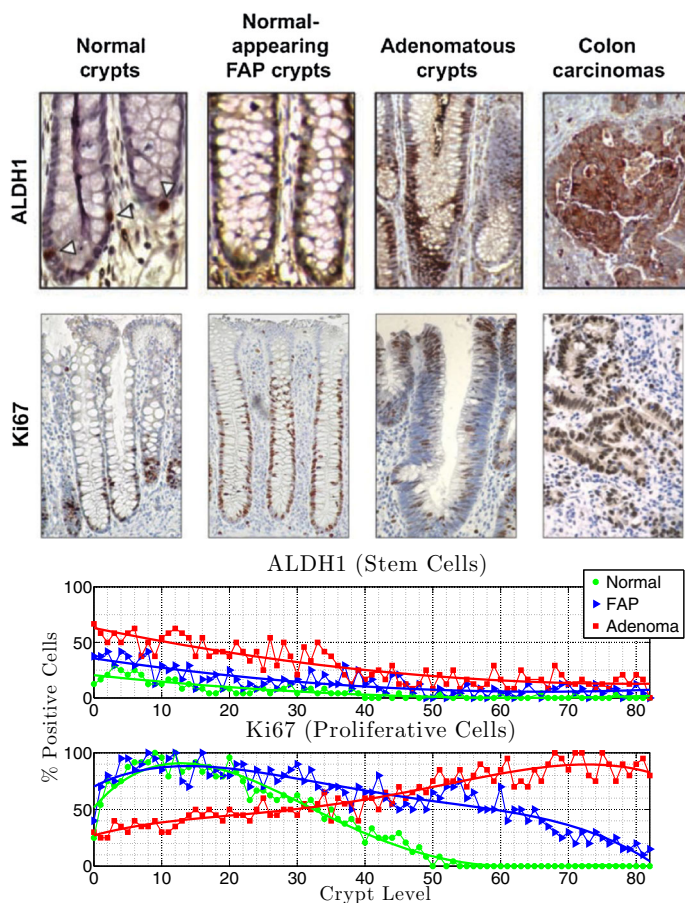


Fig. 2 First row: Immunohistochemical staining for ALDH1, an agent that detects stem cells, in normal colonic mucosa (left), normal appearing colonic mucosa from FAP patients (center left), adenomas (center right), and colon carcinomas (right). Second Row: Immunohistochemical staining for Ki-67, an agent that detects proliferative cells, in normal colonic mucosa (left), normal appearing colonic mucosa from FAP patients (center left), adenomas (center right), and colon carcinomas (right). Bottom: A plot of the percentage of cells that result in a positive detection of ALDH1 (top) and Ki67 (bottom) per crypt level, for normal, FAP, and adenoma type cell lines (Zhang et al. 2010)

15 (Potten et al. 1992). When the *APC* allele experiences the second hit mutation, the crypt becomes dysplastic and changes in tissue structure can be seen. Studies from adenomatous crypts developing from FAP patients show the LI curves are shifted even further up the crypt with the maximum toward the top (Lightdale et al. 1982; Maskens 1979). Essentially, *APC* gene mutations create a diminished APC gradient that mimics the effect of the Wnt pathway further up the the crypt thereby creating an accumulation of β -catenin within the epithelial cells. In effect, the delicate balance between cell proliferation, differentiation, and apoptosis, is disrupted creating an abundance of progenitor-like phenotypes, which can be observed in FAP patients (Boman et al. 2008). Several mapping and staining techniques were administered to confirm

an upward shift of the proliferative zone in normal-appearing FAP and adenomatous crypts (Potten and Loeffler 1990; Zhang et al. 2010; Deschner et al. 1963; Bleiberg et al. 1972; Iwana et al. 1977; Lightdale et al. 1982; Maskens 1979; Lipkin et al. 1983, 1984). In general, the transition between cell phenotypes is delayed, which shifts the population of stem-like cells toward the crypt top, eventually leading to overpopulation of proliferative cells (Boman and Fields 2013; Mills et al. 2001). In particular, Fig. 2 shows cells that stain positively for ALDH1 and Ki-67, agents that represent stem-like properties and proliferative capabilities in cells, respectively. Furthermore, Fig. 2 shows the percentage of cells that stain positively for each agent against the crypt level. We can see that in normal crypts, most stem cells and proliferative cells are contained in the bottom one-third compartment. As progressive mutations affect the crypt, the proportion of proliferative cells is larger, ultimately resulting in fully proliferative crypts and a complete loss in crypt structure.

As we can see, the canonical Wnt pathway is a complex signaling cascade that induces physiological responses essential for normal cell and tissue behavior. However, dysregulation of core pathway components such as APC and β -catenin in the colonic crypt leads to stem cell overpopulation, abnormal crypt fission and eventually colorectal cancer. We hypothesize that a molecular mechanism induces the synthesis of APC as cells divide, thereby creating inverse gradients of Wnt factors and APC along the human colonic crypt, which drives normal crypt renewal, growth, and fission. Dysregulation of these inverse gradients produce proliferative abnormalities that initiate colon cancer development. To investigate this complex process we consider mathematical modeling to allow us to explore possible biochemical and cellular mechanisms that underlie the dynamics of the crypt and development of cancer.

There has been considerable attention in the past decade on mathematical modeling both the canonical Wnt pathway and its influence in the intestine. These studies can be classified by the scale in which they are focused. A collection of models concentrate on the biochemical aspect of the cell and describe the changes in protein concentrations due to the Wnt stimulus. These models also consider the crosstalk of important components of the Wnt signal with other pathways. The cellular scale has been studied as well, where localization of proteins is modeled using compartmentalization or PDEs. Finally, at the tissue scale, individual-based and continuous models have been used to describe the cells' reactions to Wnt pathway regulation and how it affects multicellular systems. A review of these models is provided below.

The pioneering work involving a mathematical model of the canonical Wnt pathway on the biochemical scale was made by Lee et al. (2003). Lee and co-workers assume the cytosol of the cell is well-mixed so a system of non-linear, coupled ODEs can be formulated. They consider the core components of the Wnt pathway, including β -catenin, APC, and Axin, and show how these molecular concentrations change over time in response to Wnt stimulation and other factors. Lee et al. also provide parameter measurements of Wnt pathway components where possible, which were obtained from *Xenopus* oocyte extracts; however, a recent study by Tan et al. (2012) suggests that these measurements may be different compared to protein levels in mammalian cell cultures. Protein amounts, such as the total concentrations of APC, Axin, and β -catenin, were estimated from the signal intensity of Western blot analysis and from known amounts of recombinant proteins. Binding parameters were determined using

biophysical studies of interacting proteins and their specific domains. Other important parameters remain unknown, but were found by matching the degradation of β -catenin to experimental data. Lee et al. were able to validate their model by comparing the regulation of β -catenin due to changes in the scaffolding proteins APC and Axin to experimental data. They concluded that in the absence of APC, Axin becomes more involved in the destruction complex to reduce the accumulation of β -catenin, and Axin may prove useful as a target protein in therapeutic strategies.

The robustness of the Lee et al.'s model has been shown by Kruger and Heinrich (2004). The authors make several reductions in the model using conservation assumptions and rapid equilibrium approximations. Total concentrations of Dsh, GSK-3 β , APC, and TCF experience hardly any change as the Wnt pathway operates, and so the synthesis and degradations of these proteins are assumed to be negligible. With these assumptions, the model is effectively reduced from 15 equations to only seven. Kruger and Heinrich also perform a sensitivity analysis of the equilibrium concentrations to system parameters in both the Wnt stimulated and un-stimulated states. They conclude that all system variables, save for the nuclear β -catenin complex, were modulated by perturbations in parameter values. However, in the absence of Wnt, parameter variations were more likely to yield inappropriate pathway activation. A more detailed analysis of Lee's Model was considered in 2007 by Mirams et al. (2010). In this paper, a multiple time scale analysis is performed and the dominant components for each time scale are exposed. Mirams et al. show that there are three distinct time scales: one characterized by the rapid destruction of β -catenin by the destruction complex, one described by the deactivation of the destruction complex, and another that is characterized by the slower degradation of β -catenin in its free state. If one considers the dynamics on the long time scale, then the system is reduced to only one differential equation for free β -catenin. Other analyses conclude that cellular and physiological responses to the Wnt pathway may best be described by the fold-increase in β -catenin rather than changes in absolute value (Goentoro and Kirschner 2009).

Further analysis and extensions of the Lee–Heinrich model were performed by Emerick et al. (2017), which considered an important APC regulatory loop that describes the inverse relationship between APC and Wnt factors, such as β -catenin, that are typically observed in epithelial cells of the colonic crypt. Emerick et al. present two models: The first model considers an increased amount of Axin in the system, which addresses the fact illustrated by Tan et al. that protein levels observed in the *Xenopus* egg extract do not concur with levels found in mammalian cell cultures (Tan et al. 2012). The APC and Axin regulation model does not assume that the concentration of Axin is small compared to other quantities as in the Lee–Heinrich model. This provides a more accurate framework for modeling Wnt signal transduction in the human colon. The second model is a special case of APC regulation, that focuses on the interaction between APC and β -catenin and uses the parameter values and assumptions of Lee's original model (Lee et al. 2003). This model describes the dynamic interplay between APC and β -catenin that is observed in the colonic crypt and will be used extensively at the intracellular level of our multi-scale model.

We introduce the leading order system of the APC regulation derived by Emerick et al. now as it is referenced heavily in the remainder of the text. The long timescale leading order system produced by Emerick et al. is a two-dimensional system of

ODEs in APC (X_7) and β -catenin (X_{11}). The time evolution of X_{11} is also dependent on nuclear β -catenin given by X_{14} . The system is presented below in the original parameters and variable names:

$$\frac{dX_7}{dt} = \frac{v_{18}}{1 + \frac{X_{14}}{K_t} + \frac{X_{11}}{K_b}} - k_{19}X_7 \quad (1)$$

$$\frac{dX_{11}}{dt} = \frac{v_{12} - \left[k_{13} + f(W)X_7 + \frac{1}{K_{17}} \frac{dX_7}{dt} \right] X_{11}}{1 + \frac{X_7}{K_{17}}} \quad (2)$$

$$X_{14} = \frac{X_{11}TCF^0}{K_{16} + X_{11}} \quad (3)$$

The function $f(W)$ is a decreasing function of Wnt, given by

$$f(W) = \frac{W + k_2}{\alpha_1 W + \alpha_2}, \quad (4)$$

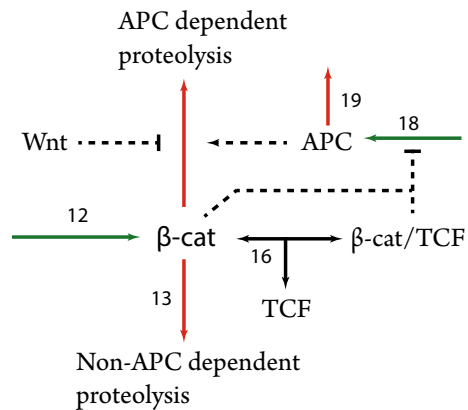
where

$$\begin{aligned} k_2 &= 1.82 \times 10^{-2} \\ \alpha_1 &= \frac{k_1 k_5 k_{15} K_7 K_8 (Dsh^0 k_3 + k_{-6})}{k_4 k_6 k_9 v_{14} GSK^0} = 697.46 \\ \alpha_2 &= \frac{k_2 k_5 k_{-6} k_{15} K_7 K_8}{k_4 k_6 k_9 v_{14} GSK^0} = 107.29. \end{aligned}$$

All parameter values except for v_{18} , k_{19} , K_t , and K_b are given by Emerick et al. (2017). To accompany these equations, we refer to the reduced schematic presented in Fig. 3, which shows the biochemical reactions between APC, β -catenin, and β -catenin/TCF and how Wnt influences the APC dependent proteolysis of β -catenin. From the figure, we see that the APC regulation model provides the desired inverse relationship between APC and β -catenin using the original parameters of the Lee–Heinrich model while implementing new parameters to be fit at the crypt level.

Tissue level dynamics are modeled using individual-based methods. Probably the most extensive model of the intestinal crypt came in 2010 by Leeuwen et al. (2010). This is a multi-scale model in which cells are identified using a Delaunay triangulation and cell division is translated by the Wnt signal to create position dependent cell proliferation and differentiation. A two dimensional plane (which represents the colonic crypt's cylindrical shape) is partitioned using a Voronoi tessellation and specific rules (i.e., when to divide, where to go after division, cell maturation and death) are assigned to each center, which may simulate a stem, transient amplifying (TA), or differentiated cell. The mechanical basis of this model is taken from Meineke et al. (2001), who assume each neighboring cell is connected via linear, over-damped springs. In this sense, when a stem cell or TA cell divides, a new cell is placed a fixed distance away from the mother cell creating a push between the surrounding cells, thus forcing them to move. Since stem cells are mostly stationary, they are modeled as such so that

Fig. 3 A reduced schematic illustrating the reaction kinetics of the reduced, long term dynamics defined by the APC Regulation model of Eqs. (1)–(3). Solid arrows depict reactions in the indicated direction (green arrows are synthesis and red arrows are destruction), double-sided arrows are binding reactions, and broken arrows denote mediated reactions (color figure online)



TA and differentiated cells migrate by mitotic pressure. At the subcellular level, this model considers Wnt pathway target genes as drivers of the cell cycle model created by Swat et al. (2004). Therefore, the cell makes decisions based on the intracellular levels of Wnt pathway components. The authors investigate several extensions to the model, including a cell–matrix adhesion and variable cell–cell adhesion, while studying the clonal expansion of the crypt, which is the process characterized by a mutant cell taking over the entire stem cell population (McDonald et al. 2006). Other multi-scale systems including Fletcher et al. (2012), Buske et al. (2011) and Mirams et al. (2012) use similar models to show that monoclonal conversion is sensitive to both the geometry of the stem cell niche and the strength of the Wnt signal in the base as well as Notch activity.

The models of tissue dynamics discussed above are more discrete in nature, where specific guidelines govern the motion and division of intestinal epithelium cells. Murray et al. worked to coarse-grain this process into a continuous model in Murray et al. (2009, 2011). The authors show strong results comparing the output of both discrete and continuous models. Furthermore, this approach is applied to the cellular dynamics of the mouse small intestine in Murray et al. (2010). Important quantitative information such as cell velocity, crypt renewal time, the total number of cells, and the fraction of proliferating cells can be generated quickly as compared to the discrete model, where many simulations must be run in order to gather similar results. In this model, biochemical dynamics are included at the cell level using a conservation of mass argument. In this sense, protein concentrations within the cell are considered as a spatial dimension, where change in these directions are assumed to be convective in nature. The authors define a cell number density given by $p(y, s, t)$, where y is distance from the bottom of the crypt, s is the subcellular concentration of β -catenin, and t is time. They apply the conservation of cell number to obtain a PDE of the form

$$\frac{\partial p}{\partial t} + \frac{\partial J_y}{\partial y} + \frac{\partial J_s}{\partial s} = r, \quad (5)$$

where J_y and J_s denote the flux of cells in y and s , respectively, and r represents a net production of cells due to cell division. Murray et al. clearly define the flux of cells in the y -direction using the coarse grained continuous model and the flux of cells in the s -direction is derived from subcellular dynamics pertaining to the Wnt pathway. The authors investigate the tissue level response to mutations at the biochemical level and determine that cell crowding caused by increased proliferation may be sufficient for monoclonal conversion of neighboring crypts.

This paper builds upon the model presented by Murray et al. (2010) by incorporating a feedback loop between APC and β -catenin at the subcellular level as proposed by Emerick et al. (2017). We apply a similar conservation of cell number idea to derive an equation such as Eq. (5) that considers a flux in both APC and β -catenin as well as physical space. The novelty of our model lies in the fact that we can investigate the effect of APC mutations at the crypt level by considering the inverse relationships between APC and β -catenin. We find that in the presence of APC mutants, the cell proliferation threshold within the crypt is shifted along the crypt axis, allowing for proliferative cells to overpopulate the crypt, which is biologically relevant.

This paper is organized as follows. In Sect. 2, we subject the underlying ODE model of Emerick et al. (2017) to a spatially varying Wnt signal. We define the crypt domain using a surface of revolution and describe the proliferative zone by degree of “stemness.” In this sense, we can examine the effects of APC mutations on the proliferative zone shifts as described by Boman and Fields (2013). In Sect. 3, we summarize the coarse grained one-dimensional cell motion model derived by Murray et al. (2009, 2010, 2011) and add a forcing term whereby APC expression is dependent on cell proliferation. We provide simulations that represent both normal and abnormal crypt dynamics. In Sect. 4, we formulate a model that includes an intracellular protein flux, and consider perturbations of this model that more accurately reflect the dynamics of the human colonic crypt at the expense of pseudo-equilibrium assumptions at the biochemical scale. In Sect. 5, we summarize the results of this paper.

2 Spatial regulation of Wnt signal transduction

We consider a spatial component to the ODE model of Emerick et al. (2017) along the major axis of the colonic crypt. Such a model gives insight into specific protein gradients that may provide information about position-dependent cell proliferation. According to Boman et al. (2010), the amount of intracellular APC increases along the crypt axis. Using immunohistochemistry and immunoprecipitation/western blotting techniques on normal colonic crypts, they were able to conclude that APC expression was decreased by 50% in the middle of the crypt and by 81% in the crypt bottom. Using pixel detection software, we were able to quantify the APC signal from several photographs of longitudinally sliced crypts (see Fig. 4). We found that in the bottom one third of the crypt, APC is negligible. However, we produce a plot of APC level by averaging that of the pixel analysis, with the maximum occurring at the top of the crypt (see Fig. 6).

The balance between Wnt-signaling components and APC homozygotes produces an optimal position dependent proliferation zone along the major crypt axis. Inside this

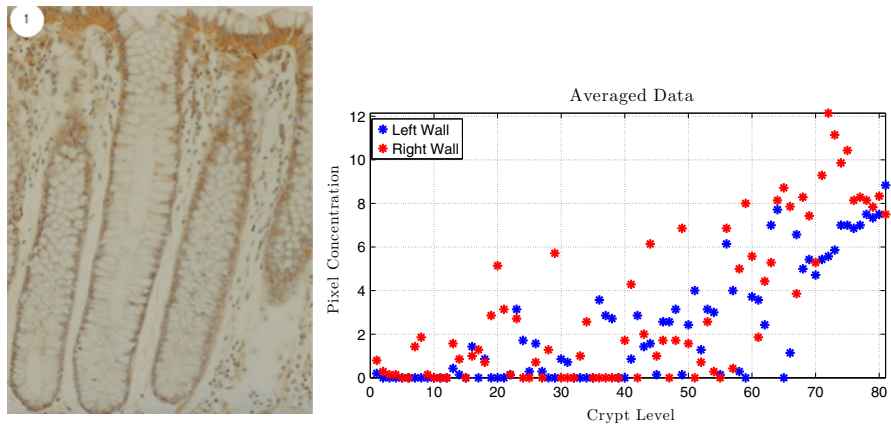


Fig. 4 (Left) Longitudinally sliced crypt provided by Midgley et al. (1997) with APC staining in brown color. (Right) Pixel analysis and plot of APC staining intensity along crypt axis (color figure online)

proliferative region exists a point of rapid proliferation, referred to as the “sweet spot.” According to Boman et al., this sweet spot occurs at approximately level 15 of 82 along the crypt axis. In the presence of APC mutations as in FAP and adenomatous crypts, we see a shift in this proliferative zone, thereby creating an abundance of proliferative cells in the crypt, which reflects their overpopulation (Boman and Fields 2013).

We formulate the spatial–temporal model and replicate the APC gradient. Then, we seek to define the proliferative zone that contains the sweet spot in normal and FAP crypts based of the level of “stemness,” which is a quantity defined using the subcellular levels of APC and downstream Wnt factors such as β -catenin.

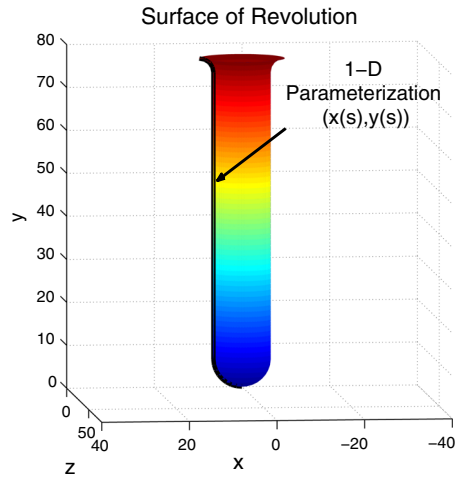
2.1 Model description and parameter fit

We expanded the simplified ODE model of Emerick et al. (2017) to a PDE model along the vertical profile of the colonic crypt. Let $(s, t) \in [0, L] \times [0, T]$, where s is a measure of individual cell positioning along the crypt, t is time, and L is the typical arc-length along the colonic crypt wall measured in cell diameters. We let $L = 82$ since the average length of the crypt is about 82 cell levels that appears as a single chain of cells stretching from the base of the crypt to the lumen of the colon. Let the parametric functions $(x(s), y(s))$ given below describe the profile of the crypt wall,

$$x(s) = \begin{cases} r_b \sin\left(\frac{s}{r_b}\right) & \text{for } 0 \leq s < \frac{\pi}{2} r_b \\ r_b & \text{for } \frac{\pi}{2} r_b \leq s < L - \frac{\pi}{2} r_t \\ r_b + r_t - r_t \sin\left(\frac{L-s}{r_t}\right) & \text{for } L - \frac{\pi}{2} r_t \leq s < L \end{cases} \quad (6)$$

$$y(s) = \begin{cases} r_b - \sqrt{r_b^2 - x(s)^2} & \text{for } 0 \leq s < \frac{\pi}{2} r_b \\ s + r_b \left(1 - \frac{\pi}{2}\right) & \text{for } \frac{\pi}{2} r_b \leq s < L - \frac{\pi}{2} r_t, \\ L - \frac{\pi}{2} (r_b + r_t) + r_b + \sqrt{r_t^2 - (x(s) - (r_b + r_t))^2} & \text{for } L - \frac{\pi}{2} r_t \leq s < L \end{cases} \quad (7)$$

Fig. 5 Crypt domain presented as a surface of revolution of the one-dimensional curve prescribed by Eqs. (6) and (7)



where $r_b = 41/2\pi$ and $r_t = 10/\pi$. The variable x is a measure of the radius of the crypt at every cell position and y is a measure of cell position along the longitudinal axis of the crypt. Here, rotating the curve about the y -axis represents an accurate depiction of the colonic crypt as the cell lineages along the crypt tend to be axially symmetric. This formulation models the crypt as a cylinder of radius r_b with a spherical cap on the bottom, also of radius r_b . The top of the crypt opens up to meet the intestinal lumen. The curvature of this opening is also modeled with a circle of radius r_t . The base of the crypt is defined by a quarter-circle whose length is 10.25 cell diameters. The crypt top is also modeled by a quarter-circle whose length is 5 cell diameters. Assigning $r_b = 41/2\pi$ and $r_t = 10/\pi$ yields a crypt wall that is 82 cell-diameters in length, which is the average number of cells from the base of a normal crypt to the top. Once we revolve this curve about the y -axis, we obtain a parameterization of the crypt with a circumference of 41 cell-diameters around the cylindrical part of the crypt. A render of our crypt domain is presented in Fig. 5.

We monitor the change in protein concentration in each cell in the context of a time-independent concentration gradient of Wnt along the crypt to represent a steady state profile of the Wnt signal. Let

$$W(s) = \exp(-y(s)/\lambda) \quad (8)$$

indicate the normalized amount of bound Wnt molecules to Frz receptors along the crypt. Here, λ is a constant value, chosen to represent a characteristic Wnt gradient. To discretize the system, we consider a chain of cells located at position $(x(s_i), y(s_i))$, for $i = 1, 2, \dots, N$. Then each cell has a constant amount of Wnt factors given by $W(s_i, t)$. Inside the cell, we denote the concentration of APC and β -catenin by $a(s, t)$ and $\beta(s, t)$, respectively, which correspond to the variables X_7 and X_{11} in the models of Emerick et al. (2017), which are given by Eqs. (1) and (2). The variables a and β represent the concentrations of APC and β -catenin, respectively, on the long timescale at leading order. There are no spatial operators so movement along the crypt wall is not

modeled, i.e., the amount of each subcellular protein concentration is only influenced by the amount of Wnt from outside the cell at the cell's location along the crypt. We consider the following system, which is the spatial analog of the APC regulation model,

$$\frac{\partial a}{\partial t} = \frac{v_{18}}{1 + \frac{TCF^0 \beta}{K_t(K_{16} + \beta)} + \frac{\beta}{K_b}} - k_{19}a \quad (9)$$

$$\frac{\partial \beta}{\partial t} = \frac{v_{12} - \left[k_{13} + f(W(s))a + \frac{1}{K_{17}} \frac{\partial a}{\partial t} \right] \beta}{1 + \frac{a}{K_{17}}}, \quad (10)$$

where f is now a function of crypt level via Wnt adapted from Eq. (4), which is provided by Emerick et al. (2017). We assume the initial conditions are constant along the crypt. That is, to solve the system we enforce the same initial conditions as the ODE system, given by

$$a(s, 0) = 97.95, \quad \beta(s, 0) = 25.10.$$

The Wnt mediated degradation of β -catenin, given by the function f , is now completely determined by the spatially varying Wnt signal. This effectively creates a β -catenin gradient along the crypt axis that is based on the input profile of Wnt.

To simulate the APC gradient, we vary the parameters v_{18} , k_{19} , K_t , and K_b of Eq. (9) as these four parameters were introduced in the APC-regulation model of Emerick et al. (2017) and have yet to be fit to spatially dependent crypt data. For the remaining parameters, we use Lee et al.'s (2003) parameter values in Eqs. (10) and (4) and we let $\lambda = 15.38$ for the Wnt signal. This choice for λ represents an accurate gradient of Wnt factors along the crypt, where the maximum is concentrated at the base. We assume that the amount of APC at the very top of the crypt is, on average, equal to the reference value of the Lee–Henrich model since this value represents the amount of baseline APC. Therefore, using the biological information provided from Zhang et al. (2010) and our quantitative analysis, we can fit the parameters to reflect an accurate APC gradient for a normal colonic crypt using a least-squares fitting approach. Considering steady state versions of Eqs. (9) and (10), we solve for β in Eq. (10), and then substitute this expression into Eq. (9) to obtain an equation of the form $F(a; v_{18}, k_{19}, K_t, K_b) = 0$, where $a = a(s)$. We then insert our biological data for a into F yielding N equations with 4 unknown parameters in each equation. We solve this system in the least-squares sense using MATLAB's built in function `lsqcurvefit`, which yields optimal parameter values for v_{18}/k_{19} , K_t and K_b . We note that since the steady state version of Eq. (9) depends on the ratio v_{18}/k_{19} , we fix $k_{19} = 1/K_{17}$ and optimize our solution in (v_{18}, K_t, K_b) space.

Figure 6 shows the spatial gradients in Wnt, β -catenin, β -catenin/TCF, and APC using the leading order Eqs. (9) and (10) with the fitted parameter values. We can also observe the effect of the Wnt signal on the protein gradients. The solutions are obtained by running the solutions of the equations until a steady state profile is reached using the built-in MATLAB ODE solver `ode15s`. APC is plotted with the biological

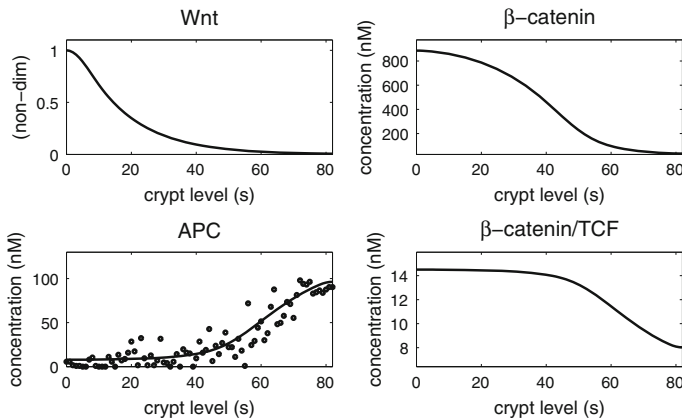


Fig. 6 Spatial gradients in β -catenin, β -catenin/TCF, and APC as determined by Eqs. (9) and (10) subject to a stationary Wnt profile, $W(s) = \exp(-y(s)/\lambda)$. The solid line denotes the fitted APC gradient to biological data (points). Parameters: $\lambda = 15.38$, $v_{18} = 0.18$, $k_{19} = 8.33 \times 10^{-4}$, $K_t = 39.21$, $K_b = 34.03$. $R^2 = 0.85$

data obtained by averaging the pixel analysis and the data from Zhang et al. (2010). Here, we take the maximum amount of APC to be that presented by Lee et al. (2003) in the reference state.

APC fits the biological profile, with its gradient opposite to those of β -catenin and nuclear β -catenin. The diminished amount of APC in the base of the crypt causes β -catenin to induce more growth as compared to the stimulated state in the Lee–Heinrich model. We note here that both parameters K_t and K_b are comparable to the levels of free and nuclear β -catenin, which indicates that APC synthesis draws equally from both the β and X_{14} (concentration of β -catenin/TCF) levels within the cell. Using least-squares, we obtain a fit with a coefficient of determination (a statistical measure of goodness of fit, one being perfect) of $R^2 = 0.85$.

2.2 Proliferative region

The spatial–temporal model gives protein gradients when subjected to a decreasing Wnt signal along the major crypt axis. The spatial profile of β -catenin is determined by the Wnt signal alone through the intracellular dynamics of the destruction complex as explained in the modified Lee–Heinrich model (2003). Because nuclear β -catenin has a positive correlation to β -catenin (Lee et al. 2003), its spatial profile is similar to that of free β -catenin. The APC gradient is influenced by Wnt-induced cell proliferation through its synthesis, which is based on an inverse relationship of both β -catenin and nuclear β -catenin. Because β -catenin is closely related to a cell's ability to proliferate and APC inactivation plays an important role in colorectal cancer, we wish to define a quantity that describes a cell's proliferative capability based on these intracellular protein levels. We continue to work under the assumption that the concentration levels of β -catenin and β -catenin/TCF indicate stem-like characteristics of a cell while an increased level of APC indicates differentiation. In this sense, we define the level of

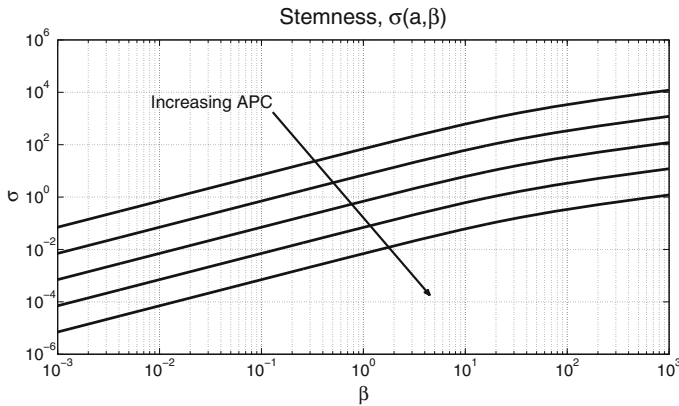


Fig. 7 A plot of the stemness function given by Eq. (11) versus β -catenin (β) for varying values of APC (a). Here, a takes on values 10^{-2} , 10^{-1} , 1, 10, and 10^2

stemness as the following dimensionless quantity:

$$\sigma := \frac{\beta \sqrt{TCF^0}}{a \sqrt{K_{16} + \beta}}. \quad (11)$$

The level of stemness is proportional to the amount of free and nuclear β -catenin since these quantities reflect a measure of the cell's proliferative potency. Stemness is inversely proportional to APC since this protein is needed for cell differentiation. Figure 7 gives a log-log plot of stemness versus β -catenin for increasing values of APC. We see that stemness increases with β -catenin but has limited increase with β -catenin when exposed to higher levels of APC. With this definition, we can move to the cellular scale to determine the optimal point along the crypt axis where cell proliferation transitions to cell differentiation. Based on the above normal profiles of nuclear β -catenin, free β -catenin, and APC, we can define a minimal stemness level, σ_p , that corresponds to the boundary of the proliferative zone along the crypt axis, $0 < s_p < L$. In normal crypts, the boundary occurs a third of the way up the crypt wall, which maintains nearly 1000 proliferative cells. For the normal case, we require $s_p = 27$, which gives a minimal stemness value of $\sigma_p = 10.42$. Figure 8 shows the normal stemness level in solid and the dashed lines correspond to a 10, 25, and 50% decrease in the production rate of APC, i.e. a decrease in the parameter v_{18} .

The proliferative zone expands to include more cells along the crypt as the production of APC diminishes. This means more cells in the crypt exhibit stem-like properties if APC production is reduced, which is precisely what happens when APC gene mutations create a loss of function in the APC protein. According to Boman et al., in FAP and adenomatous crypts, there is approximately a 25 and 50% decrease in normal APC function, respectively (Boman and Fields 2013). Here, the proliferative zone increases to more than half the crypt in FAP cases and nearly three-fourths in adenomas crypts, which is observed in the biology (Boman and Fields 2013). The increase in stemness (σ_p) due to reduced APC is consistent with biological data using the stem cell marker

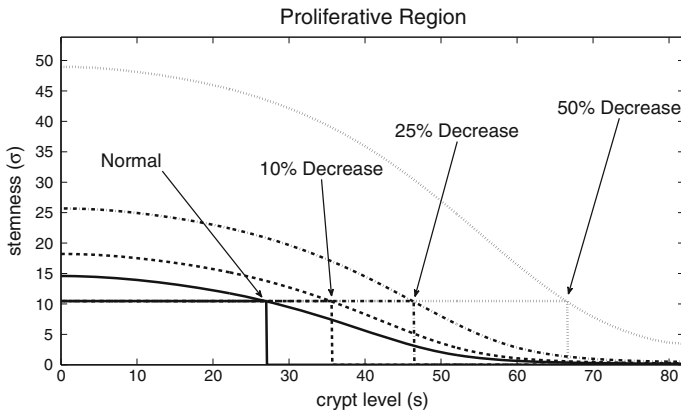


Fig. 8 Stemness values and proliferative zone shifts when subjected to diminished APC production. (Solid) Normal stemness with $\sigma_p = 10.42$ and $s_p = 27$. (Dashed) 10% decrease in v_{18} . (Dash-dot) 25% decrease in v_{18} . (Dotted) 50% decrease in v_{18}

ALDH1 showing stem cell overpopulation in FAP and adenomatous crypts (Huang et al. 2009) (see Fig. 2).

3 Cell movement and proliferation

In the previous section, we considered spatial gradients in protein levels along the crypt axis and how they are influenced by the external Wnt signal. In this sense, the cells in the crypt are in homeostasis and motion of the cells is neglected. Therefore, the model does not consider the change in intracellular protein levels as the cells move out of the Wnt signal. We require a model that tracks both the change in intracellular protein levels as determined by the external Wnt ligand while also tracking cells as they move along the crypt axis. In this way, we can more accurately determine the change in protein concentrations within each cell along the crypt.

3.1 Cell motion

A necessary first step is to define mathematically how the cells move along the crypt. Murray et al. (2009) compare a discrete and continuum model of the cell movement in the small intestinal crypt based on an off-lattice model created by Leeuwen et al. (2010) and Meineke et al. (2001). The authors use a simple assumption that a pair of neighboring cells is connected via a linear spring, which captures the fundamentals of cell elasticity and adhesion. The cells work to maintain a fixed distance, namely the resting spring length, denoted by ℓ , from each other. Through multiple simulations, Murray et al. concluded that the continuous and discrete models produce very similar results (Murray et al. 2011).

Most spatial models of the intestinal crypt are discrete and each cell is treated as an individual. Cell motion is determined by considering the forces exerted upon a given

cell by its neighbors. Leeuwen et al. (2010) and Meineke et al. (2001) implement these forces on cells that move on a fixed basal geometry, thereby neglecting the deformation of the underlying membrane. The continuous model is derived by Murray et al. by first defining a cell number density, $q(r, \tau)$, of cells per unit length at spatial coordinate r at time τ . This definition assumes that there is a large number of cells in the crypt. Cell motion is described by the following diffusive relationship:

$$\frac{\partial q}{\partial \tau} = \frac{\partial}{\partial r} \left(\frac{\alpha}{q^2} \frac{\partial q}{\partial r} \right).$$

For more details of the derivation of this equation, we refer the interested reader to Murray et al. (2009). Using this form of non-linear diffusion, we can see that a cell number density of $q = 1/\ell$, where ℓ is the resting spring length, corresponds to cells being at equilibrium and any density greater than that corresponds to spring compression, which indicates cell crowding. Furthermore, defining the non-linear diffusion coefficient as $D(q) = \alpha/q^2$, it follows that as $q \rightarrow 0$, $D \rightarrow \infty$. This implies that when the cell number density is low, the spring becomes stretched, which creates a large restoring force to bring the cells together. Conversely, if $q \rightarrow \infty$, $D \rightarrow 0$, then cell crowding creates compressed springs that are far from equilibrium. In this case, cells at the boundary of the crowd must free up space for the internal compressed cells to equilibrate. This form of non-linear diffusion, thus, describes the process of cell movement driven by cell proliferation.

For our purposes, we assume the resting spring length is $\ell = 1$, so that cells are at equilibrium when they are exactly one cell length apart. Also, our spatial coordinate is s , which corresponds to the cell level along the crypt axis, which is used as the independent variable of our spatial parameterization in Sect. 2.1. Therefore, we let $q = q(s, \tau)$ be the cell number density at spatial coordinate s along the crypt at time τ hours. We assume the crypt is spatially static, i.e., one that is already formed and is radially symmetric. The spatial scale is measured in cell diameters, approximately 10^{-5} m, and we assume $s \in [0, L]$, where $L = 82$. With this, we define

$$v_s(s, \tau) = -\frac{D(q)}{q} \frac{\partial q}{\partial s}, \quad (12)$$

where v_s denotes the cell velocity or non-linear convective flux in the s -direction. Furthermore, we assume that cells at the base of the crypt are stationary, as stem cells in the base are relatively immobile. This describes a no-flux boundary condition at $s = 0$. Finally, we assume that cells at the top of the crypt are at equilibrium, so that $q(L, \tau) = 1$. Implicit to this boundary condition is the fact that once cells reach the crypt top, they are shed into the intestinal lumen. Hence, movement into the last position is balanced by the shedding process.

3.2 Cell proliferation

We now consider a proliferation term that is coupled to the underlying intracellular dynamics. As a cell moves along the crypt wall, it receives signals from Wnt ligands

outside the cell. Cell proliferation properties are transferred via the Wnt signaling pathway, which determines the concentrations of APC and β -catenin within the cell, effectively regulating the level of stemness as defined in Sect. 2.2. In this sense, we define a proliferative region of the crypt using the Heaviside function, $H(\sigma - \sigma_p)$, where σ_p is the minimum stemness value required for cell division to occur. Cells in the proliferative region, as determined by the level of intracellular stemness, are the only cells that can divide in the crypt.

We also assume the rate of cell growth is proportional to the number of cells at each crypt level, and the cell population grows at the rate $\ln 2/T_c$, where T_c is the average doubling time of a cell in the proliferative region. This effectively describes exponential growth at each crypt level. This means cell crowding can occur within the crypt as the cells divide; however, the cells begin to push each other away as modeled by the spring forces. We obtain the following representation for the production term, denoted by R ,

$$R = H(\sigma - \sigma_p) \frac{\ln 2}{T_c} q. \quad (13)$$

Essentially, R acts as a source term in the bottom portion of the domain.

3.3 Parameter for normal cell motion and proliferation

We seek parameters that describe normal crypt dynamics in the sense of cell movement and cell reproduction in the proliferative zone. To this end, we consider a model for cell density, $q(s, \tau)$, by coupling the effects of cell motion and proliferation from the previous sections. Following the methods of Murray et al. (2011), we combine the formulations for cell motion and reproduction into a single equation involving cell number:

$$\frac{\partial q}{\partial \tau} = \frac{\partial}{\partial s} \left(\frac{\alpha}{q^2} \frac{\partial q}{\partial s} \right) + H(\sigma - \sigma_p) \frac{\ln 2}{T_c} q. \quad (14)$$

We non-dimensionalize this equation with proper scalings (see Online Appendix A) to yield the following non-dimensional parameter, γ , which is a measure of the ratio of the time scale of cell proliferation to the time scale of cell movement. This parameter is given by

$$\gamma = \frac{L^2 \ln 2}{\ell^2 \alpha T_c}. \quad (15)$$

Murray et al. derive a characteristic value for γ for the murine small intestinal crypt (Murray et al. 2011) based on the total number of cells in the crypt, the number of proliferative cells, and the renewal time, i.e. the time it takes for the cell lining of the crypt to turnover completely. Using these three factors, they were able to conclude a γ value of 0.30, which is due to the short crypt size and lower cell count. We use a similar derivation in Online Appendix A to conclude that $\gamma = 0.17$ for the normal large intestinal crypt.

4 Intracellular protein flux

The evolution of the cell density $q(s, \tau)$ captures the dynamics of a cell population moving along the crypt wall as more cells are created in the base. Furthermore, we can acquire values for γ and σ_p that correspond to normal crypt properties. Studies by Roberts et al. (1995) suggest that stemness is lost gradually, and eventually stem cells lose the capacity to function as stem cells. Therefore, we seek to define a system where the cell's protein levels change dynamically with the cell's spatial position. To this end, we implement additional spatial dimensions that reflect the stemness of the cell. We formulate the model below assuming that intracellular dynamics are given by the APC regulation model of Emerick et al. (2017).

4.1 APC and β -catenin as spatial dimensions

We consider a cell density $p = p(s, a, \beta, \tau)$, where p denotes the number of cells per unit arc-length s , with intracellular protein concentrations of a and β , where a denotes the level of APC and β represents the amount of β -catenin, at time τ . Here, $s \in E_s = [0, L]$ is the distance along the the major crypt axis, where L denotes the typical arc-length of a crypt in cell diameters. The changes in protein concentrations inside the cell are determined by the APC regulation model, and are incorporated into the evolution equation by means of a convective velocity. In this sense, the biochemical protein concentrations act as an “age” dynamic for the cell, which mathematically is modeled as a spatial dimension. It follows that the variables a and β originate from the intervals $E_a = [0, a_{max}]$ and $E_\beta = [0, \beta_{max}]$, respectively, as any non-negative protein amount up to a maximum can be considered. Furthermore, the quantity of cells per unit arc-length is given by the following integral

$$q(s, \tau) = \int_0^{a_{max}} \int_0^{\beta_{max}} p(s, a, \beta, \tau) da d\beta. \quad (16)$$

This quantity is defined as the total number of cells at position s along the crypt at time τ , which is identical to the cell number density considered in the last section. We wish to monitor the change in p at time $\tau > 0$. To construct the evolutionary PDE, we consider a control volume, Ω , in (s, a, β) space, then the conservation law assumes

$$\frac{d}{d\tau} \int_{\Omega} p dV = - \int_{\partial\Omega} \mathbf{J} \cdot d\mathbf{S} + \int_{\Omega} R dV, \quad (17)$$

where $\mathbf{J} = \langle J_s, J_a, J_\beta \rangle$ is a three-dimensional vector that describes the flux of cells in each direction through the surface $\partial\Omega$ and R is a net production of cells inside the arbitrary volume Ω . We consider the balance of fluxes and net production, R , inside Ω on the quantity p . Using the divergence theorem on Eq. (17), and assuming the continuity of p , we obtain the following

$$\int_{\Omega} \frac{\partial p}{\partial \tau} dV = \int_{\Omega} [-\nabla \cdot \mathbf{J} + R] dV,$$

where $\nabla = \left\langle \frac{\partial}{\partial s}, \frac{\partial}{\partial a}, \frac{\partial}{\partial \beta} \right\rangle$. Since we are considering an arbitrary volume under the usual smoothness assumptions, we obtain the following:

$$\frac{\partial p}{\partial \tau} + \frac{\partial}{\partial s} J_s + \frac{\partial}{\partial a} J_a + \frac{\partial}{\partial \beta} J_\beta = R, \quad (18)$$

which holds for all $(s, a, \beta) \in E = E_s \times E_a \times E_\beta$ and for all $\tau > 0$. As in the previous section, we use the convective flux in the s -direction, $J_s = v_s p$, where v_s is the cell velocity given by Eq. (12). We apply a no-flux boundary condition on the quantity $q(s, \tau)$ at $s = 0$ and the equilibrium condition $q(L, \tau) = 1$ at the top of the crypt. Furthermore, the source term R is given by Eq. (13), except the dependence is on p rather than q .

Following the work of Murray et al. (2010), we assume the flux in both β -catenin and APC is convective with a diffusive element that describes any random fluctuation of proteins that may occur between cells at the cellular scale. Here, the diffusion is a measure of the noise at the population scale that may result from intracellular variability in Wnt signaling. In this way, the fluxes are given by

$$J_a = v_a p - D_a \frac{\partial p}{\partial a}, \quad J_\beta = v_\beta p - D_\beta \frac{\partial p}{\partial \beta} \quad (19)$$

where v_a, v_β are the convective velocities and D_a, D_β are the constant diffusion coefficients. We incorporate the underlying ODE dynamics given by the APC regulation model by defining v_a and v_β by Eqs. (9) and (10), respectively, given by

$$v_a = \frac{v_{18}}{1 + \frac{TCF^0 \beta}{K_t(K_{16} + \beta)} + \frac{\beta}{K_b}} - k_{19}a. \quad (20)$$

$$v_\beta = \frac{v_{12} - \left[k_{13} + \left(\frac{W + k_2}{\alpha_1 W + \alpha_2} \right) a + \frac{1}{K_{17}} v_a \right] \beta}{1 + \frac{a}{K_{17}}}. \quad (21)$$

We note here that the above equations evolve on the t time scale, and hence must be appropriately scaled to the τ time scale in which the PDE evolves. In this way, we can monitor the change in subcellular protein levels while tracking the cells as they move up the crypt and out of the range of the Wnt signal. We note here that the Wnt signal is a spatially varying, time independent quantity, $W = W(s)$. We employ no-flux boundary conditions at $a = 0$ and $a = a_{max}$, and $\beta = 0$ and $\beta = \beta_{max}$, where a_{max} and β_{max} are upper bounds for non-dimensional APC and β -catenin levels, since no cells can obtain a negative protein value or go above the maximum value of protein concentration as allowed by the ODE model. Using convective velocities described above, we can investigate the changes at the cellular level due to fluctuations in APC production, which effectively mimics APC mutations. Furthermore, the multiple time scale analysis performed by Emerick et al. (2017) allows us to restrict our attention to dimensions in APC and β -catenin only, which reduces computational costs and simplifies the model tremendously. This facilitates fitting to baseline biologic data on

APC and β -catenin. Indeed, we have experimental data available on APC (see Fig. 4) for normal crypts and the concentration of nuclear β -catenin decreases inversely to the increase of APC along the crypt axis (van de Wetering et al. 2002). Lastly, we remark that the conditions at $s = 0$ and $s = L$ are defined in terms of cell density given by $q(s, \tau)$ from Eq. (16), which means the conditions are non-local boundary conditions. The details concerning well-posedness are not known; however, numerical computations of the system converge to a unique solution depending on initial data.

4.2 Normal crypt dynamics

In this section, we investigate the numerical results of a reduced model. We assume the APC dynamics are in pseudo-steady state equilibrium. We assume there is no-flux in the a -direction so that $J_a = 0$. This implies $v_a = 0$ and $D_a = 0$, so that in Eq. (20) we get

$$a = \frac{v_{18}/k_{19}}{1 + \frac{TCF^0\beta}{K_t(K_{16}+\beta)} + \frac{\beta}{K_b}}. \quad (22)$$

This means the amount of APC inside the cell is immediately determined by the amount of free and nuclear β -catenin within the cell. In this sense, the convective velocity in the β direction is now given by

$$v_\beta = \frac{v_{12} - \left[k_{13} + \left(\frac{W+k_2}{\alpha_1 W + \alpha_2} \right) \left(\frac{v_{18}/k_{19}}{1 + \frac{TCF^0\beta}{K_t(K_{16}+\beta)} + \frac{\beta}{K_b}} \right) \right] \beta}{1 + \frac{v_{18}/(k_{19}K_{17})}{1 + \frac{TCF^0\beta}{K_t(K_{16}+\beta)} + \frac{\beta}{K_b}}}. \quad (23)$$

The cell number density is a function of arc-length, β -catenin, and time. Hence, Eq. (18) can be written as

$$\frac{\partial p}{\partial \tau} = \frac{\partial}{\partial s} \left[\frac{\alpha}{q^3} \frac{\partial q}{\partial s} p \right] - \frac{\partial}{\partial \beta} \left[v_\beta p - D_\beta \frac{\partial p}{\partial \beta} \right] + H(\sigma - \sigma_p) \frac{\ln 2}{T_c} p, \quad (24)$$

subject to the non-local boundary conditions in s given by Eqs. (A.2) and (A.3), and no-flux boundary conditions in β . Similar to the model in Sect. 3.2, we non-dimensionalize Eq. (24) according to the following scalings:

$$\hat{p} = p, \quad \hat{q} = \ell q, \quad \hat{\tau} = \frac{\ln 2}{T_c} \tau, \quad \hat{s} = \frac{s}{L}, \quad \hat{\beta} = \sqrt{\frac{\ln 2}{D_\beta T_c}} \beta, \quad \hat{v}_\beta = \frac{v_\beta}{v_{12}}.$$

Upon substitution of the scaled variables and dropping the hat notation for convenience, we obtain the following

$$\left\{ \begin{array}{l} \frac{\partial p}{\partial \tau} = \frac{1}{\gamma} \frac{\partial}{\partial s} \left[\frac{1}{q^3} \frac{\partial q}{\partial s} p \right] - \frac{1}{\eta} \frac{\partial}{\partial \beta} (v_{\beta} p) \\ \quad + \frac{\partial^2 p}{\partial \beta^2} + H(\sigma - \sigma_p) p \quad (s, \beta) \in (0, 1) \times (0, \hat{\beta}_{max}), \tau > 0 \\ \frac{\partial q}{\partial s} (0, \tau) = 0 \quad \text{for } \tau > 0 \\ q(1, \tau) = 1 \quad \text{for } \tau > 0 \\ J_{\beta}(s, 0, \tau) = 0 \quad \text{for } s \in (0, 1), \tau > 0 \\ J_{\beta}(s, \hat{\beta}_{max}, \tau) = 0 \quad \text{for } s \in (0, 1), \tau > 0 \\ p(s, \beta, 0) = p_0 \quad \text{for } (s, \beta) \in (0, 1) \times (0, \hat{\beta}_{max}), \end{array} \right. \quad \begin{array}{l} (25) \\ \\ (26) \\ (27) \\ (28) \\ (29) \\ (30) \end{array}$$

where the non-dimensional parameters γ and η are given by

$$\gamma = \frac{L^2 \ln 2}{\ell^2 \alpha T_c}, \quad \eta = \sqrt{\frac{D_{\beta} \ln 2}{v_{12}^2 T_c}}. \quad (31)$$

The parameter γ measures the ratio of the time scale of cell proliferation to cell motion and η measures the effects of proliferation, convection, and diffusion in the β -direction. Here, γ is the same as in the last section, and we keep it fixed at $\gamma = 0.17$. We wish to determine the effects of the parameter η on the output of the model. From the equation, it is apparent that η controls the effect of diffusion on the convective velocity, v_{β} . Considering the case when $\eta \gg 1$, we can assume the diffusion coefficient is large. Hence, the convective velocity is diminished by the linear diffusion term. This will smooth the cell number density in the β -direction, resulting in a high variability of stemness at the cellular level. In contrast, if $\eta \ll 1$, then $D_{\beta} \ll 1$ and diffusion has less impact. The convective velocity v_{β} controls the flux in β -space, yielding a rigid profile in the shape of $v_{\beta} = 0$. To get a sense of the effects of η on the steady state profile of cell number density, we plot the result of Eq. (25) for small and large values of η in Fig. 9. We run a finite difference scheme in the spatial dimensions and use the built-in MATLAB solver `ode15s` to step in time (see Online Appendix B). We denote the average amount of β -catenin per crypt level as the following quantity

$$A_{\beta}(s, \tau) = \frac{1}{q(s, \tau)} \int_0^{\beta_{max}} \beta p(s, \beta, \tau) d\beta.$$

Figure 9 shows how diffusion alters the steady state surface of the cell number density, p , in (s, β) space. In the left plot, $D_{\beta} = 15$, which gives a thinner profile that adheres closely to the zero contour of v_{β} . This indicates a smaller variability of subcellular protein levels at the cellular scale. In contrast, the right plot shows the steady state cell number density for $D_{\beta} = 200$, which smooths the data in the β -direction. This leads to a higher variability at the cellular scale. Depending on this

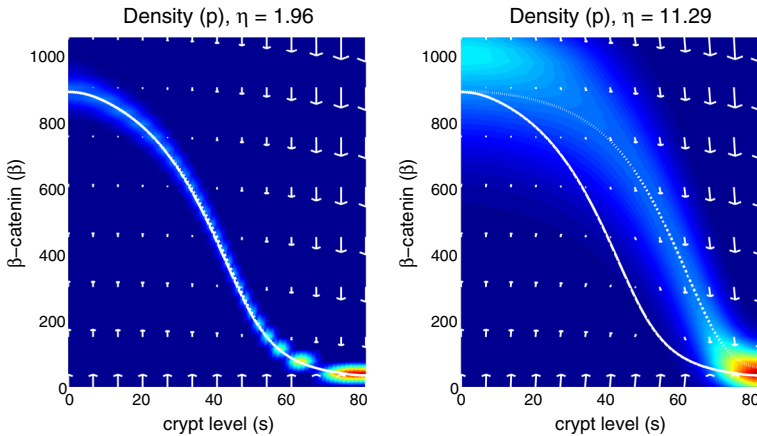


Fig. 9 Surface plot of $p(s, \beta, \tau)$ at steady state from Eqs. (25)–(30) with varying η . The solid line shows the zero contour of the convective velocity, $v_\beta = 0$, and the average amount of β -catenin, A_β , is shown as the dashed-dotted line. The arrows denote the cell velocity field, (v_s, v_β) . The color scheme represents the value of the cell density, p (dark blue = 0, red = max p). Parameters: $\gamma = 0.17$, $\sigma_p = 16.42$, (left) $D_\beta = 15$, (right) $D_\beta = 200$. All parameters for v_β are given by Emerick et al. (2017). Numerical scheme for solutions is detailed in Online Appendix B (color figure online)

variability, there may be more or less cells in the proliferative zone at any given time. We can see that when $D_\beta = 200$, the cell density profile and the average amount of β -catenin is slightly shifted to the right as compared to the zero contour of v_β . This is due to the proliferative zone consisting of more cells, which ultimately creates a larger cell velocity in the s -direction, v_s . Indeed, for large η , we can see that the cell velocity field contains more of an s -component. Therefore, η not only controls the protein variability but also the amount of cells, renewal time, and the average protein levels inside the cell along the crypt axis.

Furthermore, from Fig. 9, we can see that p achieves the desired profile in both cases. Most of the cells in the base of the crypt have a much higher level of β -catenin since the Wnt signal is the strongest there. As cells migrate up the crypt, the levels of intracellular β -catenin drop since cells are moving out of the Wnt signal and APC is increasing due to increasing cell division. However, we can see that the parameter η distorts the β -catenin gradient, resulting in a shifted APC gradient (see Fig. 10). Therefore, we must be careful in choosing the parameter value for η since the average APC per crypt level will deviate from our data. Also, Figure 9 shows steady state solutions for $\sigma_p = 16.42$, which is a higher stemness threshold than before, but still yields comparable solutions for the APC profile. However, the number of proliferative cells in the crypt at steady state is extremely low due to the high stemness threshold. We seek feasible parameter values for η and σ_p , while maintaining a desirable APC gradient that results in normal crypt dynamics.

To this end, we perform an exhaustive search in (η, σ_p) space in a similar way that we found the parameters for γ and s_p in the reduced model (see Fig. 2). That is, we initially find a feasible parameter set that corresponds to the normal total number of cells, total number of proliferating cells, and renewal time. Then, we fix η and σ_p

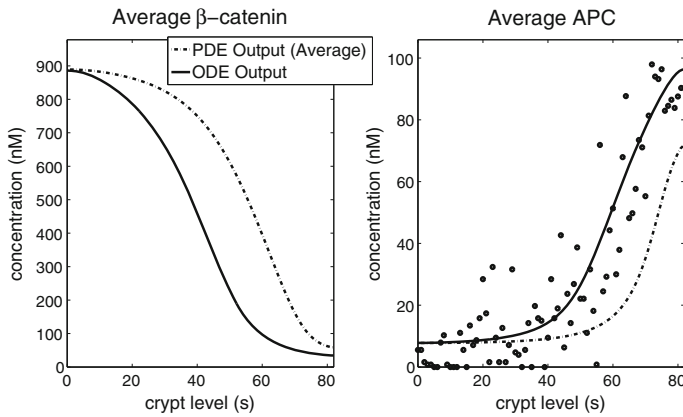


Fig. 10 (Solid) Steady state output β -catenin and APC per crypt level of underlying PDE model given by Eqs. (9) and (10). (Dashed-dotted) Average amounts of β -catenin [given by $A_\beta(s, \tau)$] and APC [given by Eq. (22)] per crypt level after running Eqs. (25)–(30) to steady state. Parameters for PDE: $\gamma = 0.17$, $\sigma_p = 16.42$, $\eta = 11.29$

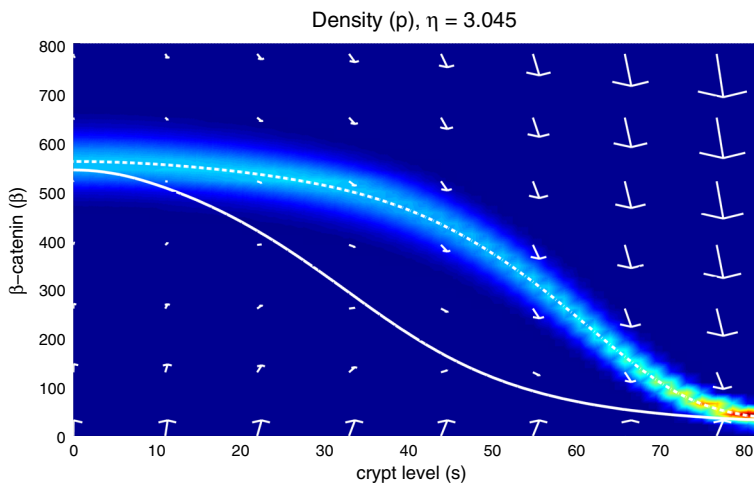


Fig. 11 Surface plot of $p(s, \beta, \tau)$ at steady state from Eqs. (25)–(30) with optimal values for η , σ_p , v_{18} , K_t , and K_b . The solid white line shows the zero contour of the convective velocity, $v_\beta = 0$ and the average β -catenin, A_β , is shown by the white dashed-dotted line. The arrows denote the velocity field, $\langle v_s, v_\beta \rangle$. Parameters: $\gamma = 0.17$, $\eta = 3.05$, $\sigma_p = 6.52$, $k_{19} = 1/K_{17}$, $v_{18} = 0.17$, $K_t = 18.25$, and $K_b = 60.44$. Numerical scheme for solutions is detailed in Online Appendix B

and update the parameters v_{18} , K_t , and K_b that yield a relevant biological fit in the non-linear least squares sense as in Sect. 2.1. We continue this iterative process until an optimal parameter set is found. Figure 11 shows the steady state profile of $p(s, \beta, \tau)$ in (s, β) space. To obtain normal crypt dynamics, we notice that the maximum amount of average β -catenin (dashed-dotted white line) in the base of the crypt is more than one-third less than the output of the PDE model of Sect. 2.1 (see Fig. 6). Therefore, the parameter set for the underlying ODE dynamics are set to achieve the desired drop in

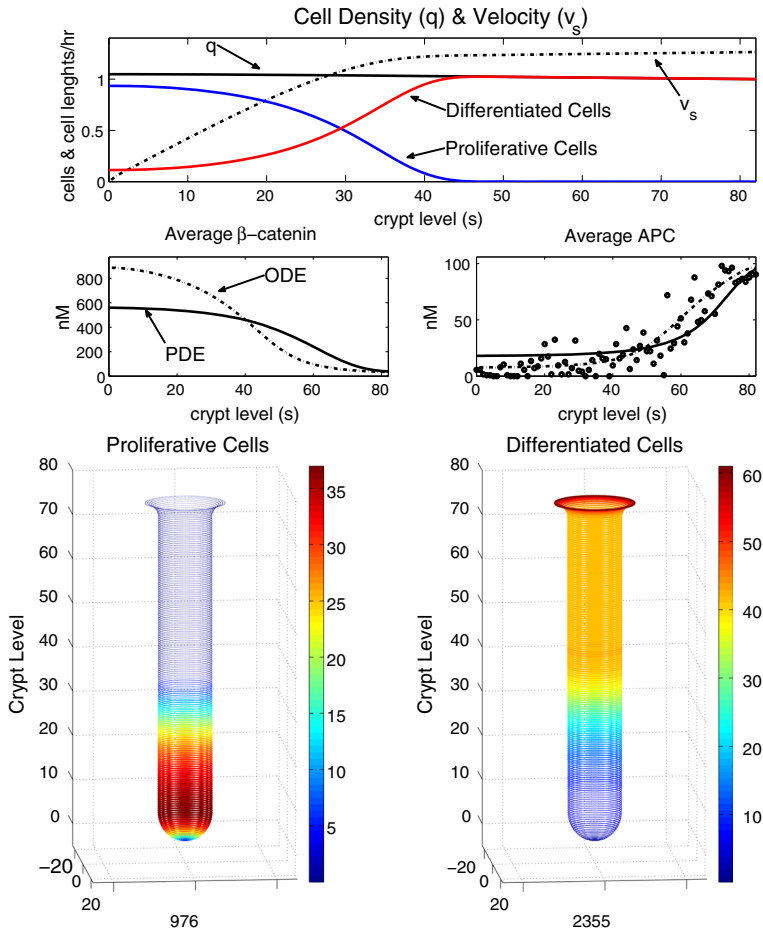


Fig. 12 (Top) Plot of q and v_s at steady state. (Middle) Plots of average β -catenin and APC from Eqs. (25)–(30) (solid lines) and output from the spatial ODE model of Sect. 2.1 (dashed dotted lines). (Bottom) Steady state profiles of proliferative and differentiated cells plotted on the full crypt domain. Same parameters as Fig. 11. Numerical scheme for solutions is detailed in Online Appendix B

stemness as cells move up the crypt that gives optimal values for renewal time and total number of cells. This results in a slight decrease in β -catenin concentration, which means there is more APC in the base of the crypt than originally expected (see Fig. 12). However, this decrease in β -catenin is actually more comparable to Lee's original model (Lee et al. 2003). We can also see that setting $\eta = 3.05$ results in a diffusion coefficient of $D_b = 36.37$, which yields a shifted density profile (as compared to the zero contour, $v_b = 0$) at steady state. This shift occurs because the smaller β -catenin concentration coupled with the relatively high diffusion means more cells are within the proliferative zone, which creates a faster cell velocity, v_s . Therefore, as cells are moving out of the region of high Wnt signal, the cells' intracellular levels of β -catenin change based on their position. The speed of migration is determined by v_β , which

in the normal case is comparable to v_s as cells move up the crypt. Ultimately, we arrive at a balance that produces the velocity field and steady state cell density surface illustrated in Fig. 11.

From this output, we can also determine the distribution of each cell type and the average amounts of protein concentrations versus crypt level. Figure 12, shows the steady state distribution of proliferative cells and differentiated cells. We can see that most of the proliferative cells are in the crypt base and as they move up the crypt, they transition smoothly to form differentiated cells. This transition is based on the level of stemness of the cell and mimics the progression from proliferative cells to differentiated cells that occurs naturally in the crypt (see Fig. 2). Further, the figure shows the average amount of both β -catenin and APC versus crypt height compared to the model presented in Sect. 2.1. Indeed, we can see that the PDE gives a smaller value of β -catenin in the base compared to our simplified model. The amount of APC is elevated in the base of the crypt, but achieves the normal height at the crypt top. This change from the PDE model in Eqs. (9) and (10) is necessary to maintain normal crypt dynamics while still achieving a high goodness of fit value (R^2) as compared to our data, which in this case is $R^2 = .77$. Finally, Fig. 12 also shows the crypt layout with the number of proliferative cells totaling 976 out of a total 3331 cells. This means the crypt has 29% proliferative cells which falls in the normal zone. Also, the renewal time for the normal parameter set is 5.03 days.

4.3 Abnormal crypt dynamics

We perturb the parameter v_{18} in the underlying, simplified APC regulation model to investigate the behavior of the current model in scenarios that simulate colorectal cancer development. As in the previous sections, we let 25 and 50% decreases in APC production mimic the mutation process in FAP and adenomatous crypts, respectively. A diminished APC gradient is imposed, which causes more cells to become proliferative as defined by the intracellular level of stemness.

The results for each scenario, along with a 10% decrease in APC production, are shown in Figs. 13 and 14. These figures show the density (p) at steady state in (s, σ) space along with the cell type profiles along crypt level and the density plot on the full crypt domain to illustrate how many proliferative cells are at each level of the crypt. We note that in the normal case, less than half of the cell number density is above the proliferative zone threshold. However, with only a 10% decrease in the parameter describing APC synthesis (v_{18}), we see that over half of the cell density is above the proliferative threshold. This results in 57% of the cells being proliferative, which can be seen from the cell type profile and the full crypt plot in Fig. 13. The steady state density is almost completely inside the proliferative zone when subjected to a 25% decrease in APC production. This yields a total of 3019 proliferative cells, which almost equates to a completely proliferative crypt (88% proliferative). With a 50% decrease in v_{18} , we see that the cell density is completely above the threshold and that all of the cells have nearly twice the level of stemness of the normal case. Furthermore, we can see that in simulating FAP and adenomas, the velocity of the cells reaches almost 4 cell lengths/h by the time the cell gets to the top of the crypt. This

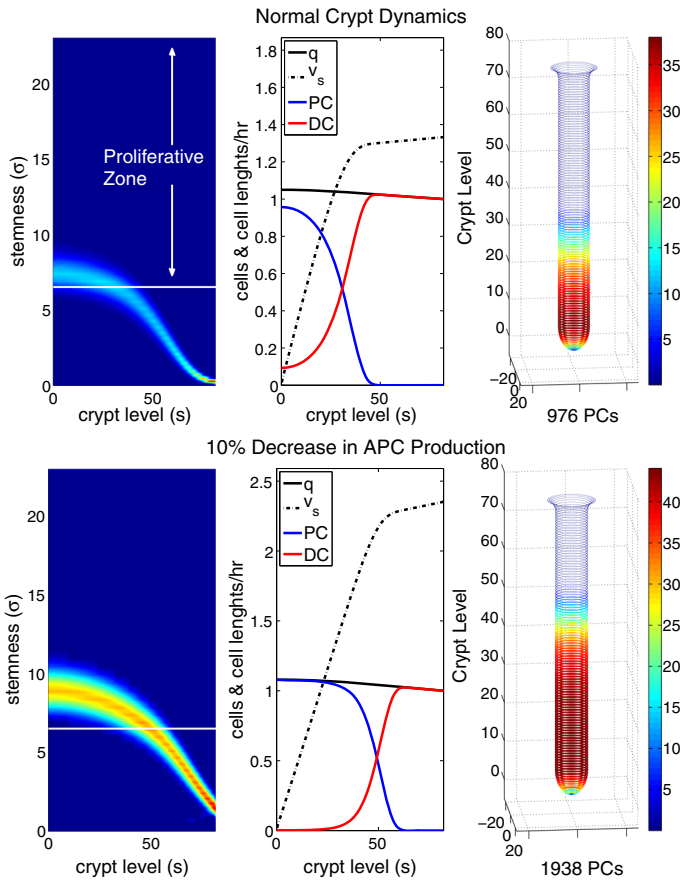


Fig. 13 A surface plot of the cell number density p at steady state in (s, σ) space (the white line is $\sigma = \sigma_p$), a plot of q , v_s and cell-type curves (PC = proliferative cells, DC = differentiated cells) against crypt level, and the density of proliferative cells plotted on the full crypt domain. (Top) Normal crypt dynamics. (Bottom) 10% drop in v_{18} . All other parameters are the same as in Fig. 11. Numerical scheme for solutions is detailed in Online Appendix B

is more than double the normal case. Because of this, the crypt renewal time drops to about 3.9 days in each case compared to a 5.03 day renewal time in the normal case. Finally, in the adenoma case, we see an increase of only 113 cells compared to the normal, which is comparable to the results from the more simplified models of the previous sections.

We conclude that the system presented by Eqs. (25)–(30) is sensitive to intracellular perturbations in APC production. The inclusion of the β -catenin flux prescribed by v_β causes a slight delay in the Wnt signal transduction by each cell in the crypt. This, essentially, causes more cells to be in the proliferative zone as they move up the crypt; thereby creating a more APC responsive system. Hence, any decrease in APC will have a more devastating effect on the multi-cellular level. We note here that in the case of a 25% drop in v_{18} , the cell layout is comparable to the normal appearing FAP crypt

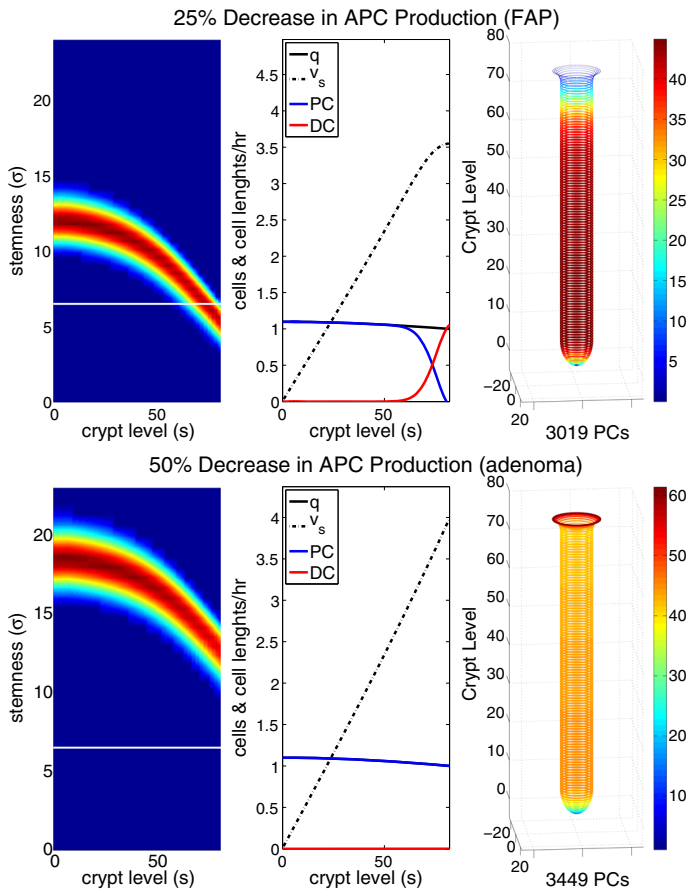


Fig. 14 A surface plot of the cell number density p at steady state in (s, σ) space (the white line is $\sigma = \sigma_p$), a plot of q , v_s and cell-type curves (PC = proliferative cells, DC = differentiated cells) against crypt level, and the density of proliferative cells plotted on the full crypt domain. (Top) 25% drop in v_{18} (FAP). (Bottom) 50% drop in v_{18} (adenoma). All other parameters are the same as in Fig. 11. Numerical scheme for solutions is detailed in Online Appendix B

as presented in Zhang et al. (2010) and in the adenoma case, we obtain a completely proliferative crypt that compares favorably with the results of Boman et al. (2010) (see Fig. 2).

5 Discussion

The goal of this paper was to effectively model crypt level dynamics in the human colon by implementing inverse profiles of APC and Wnt factors that ultimately determine proliferative capabilities of the cell. In this sense, normal and abnormal crypt dynamics are manipulated by parameters involving Wnt signal transduction and we can simulate the behavior of cell populations in the crypt in the presence of APC mutations.

The main contribution of this paper is the crypt modeling framework as it applies to the human colon. We used a multi-scale modeling approach that takes into consideration the intracellular protein dynamics of the Wnt pathway, which effectively defines stemness within the cell. In Sect. 2, we fit the APC regulation parameters of Emerick et al. (2017) to a spatially varying protein gradient found by pixel detection analysis. We were able to fit cell motion and proliferation parameters to specific colonic crypt dynamics in Sect. 3, which gives a platform for subjecting the system to simulate APC mutations. Once this was accomplished, we were able to use the more complex system given by Eqs. (25)–(30) to fit protein velocity parameters to normal crypt dynamics. Section 4 presents a specific technique to capture the spatial movement of cells as prescribed by Leeuwen et al. (2010) and the intracellular protein flux as described by the APC regulation model of Emerick et al. (2017). Using this framework, we are able to capture the dynamics of the colonic crypt using a transition from protein concentrations to cellular proliferation based on the strength of opposing protein gradients along the crypt as observed by Boman and Fields (2013).

In colon cancer, the existence of crypt structures is completely absent suggesting that overpopulation ultimately destroys normal crypt function and geometry (Boman and Fields 2013). Such a phenomenon may be analogous to our model when $\gamma \gg 1$. In this case, proliferation overtakes migration, trapping cells in the crypt. Because there is limited space in the crypt, the cells must move perpendicular to the crypt wall. Our model cannot capture this physical property since it only considers one spatial dimension. However, in Sect. 3.3, as s_p (the spatial coordinate along the crypt wall that corresponds to the stemness threshold, σ_p) approaches L , representing a fully proliferative crypt, the γ -value of 0.17 is still small enough to ensure the cells do not exponentially increase within the crypt. This suggests that a more significant increase of cells may result in buckling of the primary epithelium layer leading to abnormal crypt fission, which is ultimately the cause of aberrant crypt geometry. However, to capture this dynamic, future work may include a higher dimensional representation of the crypt domain, where domain growth is coupled to the cell number density.

In addition to investigating the causes of colon cancer development, it is important to discuss the parameters that are responsible for bringing the crypt back to a normal state when the APC gradient is diminished. Typically in neoplastic crypts, it would be desirable to establish more functional APC to essentially reset the amount of β -catenin within the cell. Therefore, it is beneficial to biologists to determine what other Wnt pathway components could potentially be targeted to counteract the decrease in functional APC. One measure that we consider that determines whether a crypt is abnormal or normal is the total amount of stemness in the crypt, since typically the stemness value of the cells will determine the total number of cells and the crypt's proliferation at steady state. The total stemness is calculated by

$$[\text{Total Stemness}] = \int_0^L \sigma(s) ds,$$

where σ is the steady state profile of stemness, which is calculated in Sect. 2.2 as the ratio of Wnt factors to APC. We consider parameter perturbations in the APC regulation model in the form $k_i^* = (1 + \delta)k_i$, where k_i^* is the value needed to bring the

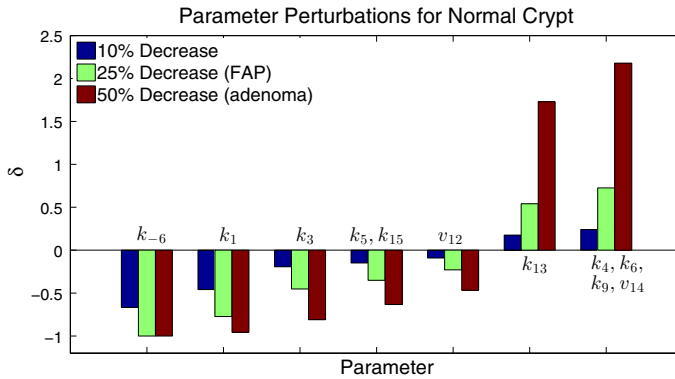


Fig. 15 Bar graph illustrating the increase or decrease needed in each parameter of the APC regulation model to restore the total amount of stemness back to normal in the presence of a 10% (blue), 25% (green), and 50% (red) decrease in APC production (v_{18}). The height of each bar is δ from the formula $k_i^* = (1 + \delta)k_i$, where k_i^* is the parameter needed in order to recover the normal stemness and k_i is the original parameter value from Emerick et al. (2017) (color figure online)

stemness of the crypt back down to the normal value in the presence of APC mutations. Here, k_i is any parameter from the APC regulation model of Emerick et al. (2017) that represents a synthesis, degradation, or binding rate except for reduced APC regulation model parameters and dissociation constants. Also, δ measures the amount at which we have to deviate from the original parameter value, which is bounded below by -1 since $k_i^* > 0$. Figure 15 shows a bar graph of perturbation values (δ) needed to establish the normal total stemness in the crypt in the presence of a 10%, 25%, and 50% drop in APC production. Here, we can see that total stemness is the most sensitive to the synthesis rate of β -catenin, v_{12} , since it only requires about a 50% drop in v_{12} to revert back to normal under the adenoma scenario.

In contrast, the mediated release of GSK-3 β from the destruction complex (k_3) and the Wnt mediated switch from inactive to active Dishevelled (k_1) must be almost completely turned off in order to restore the crypt's stemness in the adenoma case. This means one must completely deactivate Dishevelled's influence on the destruction complex or block the Frizzled receptor from binding to a Wnt ligand. Furthermore, we note that in the simulated FAP and adenoma cases, the parameter k_{-6} , which represents the dissociation rate of GSK-3 β from APC/Axin, cannot be reduced enough to reset the system. Total stemness is more sensitive to reductions in the rate of de-phosphorylation of the destruction complex (k_5) and the rate of Axin degradation (k_{15}). These rates have to be reduced by more than half to restore the system in the adenoma case. We can also see from the figure that some parameters must be increased in order to restore normal crypt stemness. In the FAP case, the rate of β -catenin degradation (k_{13}) has to be increased by 50% and in the adenoma case it has to be almost tripled to return the system to normal. The remaining parameters act similarly, and must be more than tripled in order for the normal stemness value to return. Similar control analyses can be done in order to test the intracellular flux model's sensitivity to underlying APC regulation parameters. Thus, of the factors that would revert decreased APC states

back to normal, our modeling indicates that targeting components of the Wnt pathway that contain Axin would be most effective.

We make a final note about the future directions of the models developed in this paper. One area that we would expand on in the future is to implement a stem cell and transient amplifying (TA) cell population into the model. This means there would essentially be two stemness threshold values that determine the proliferative capabilities of both the stem cells and TA cells. We would like to also implement stem cell and TA cell division rates with a probability of self renewal as described in recent models created by Nakata et al. (2012) and Rodriguez-Brenes et al. (2013). Furthermore, we would like to investigate an APC feedback mechanism that is controlled by an element at the tissue scale such as the number of times a cell has divided. In this way, we can directly relate cell divisions to the amount of intracellular APC. This means our convective velocity v_β [see Eq. (25)] is a function of the total cell density, p , and so changes at the multi-cell scale will have an influence on the intracellular protein levels.

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